

Particle-based and stochastic traffic assignment

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1 Short summary

The particle-based and stochastic traffic assignment problem is considered. The increasing deployment of “agent-based” traffic assignment models in transport planning renders this problem relevant. Lacking a comprehensive mathematical characterization, the computation of an assignment within this model class is bound to heuristics. One of these approaches is the “sorting technique” of Sbayti et al. (2007). The present article offers a derivation of this method from first principles, works out the effect of model stochasticity, and uses these results to propose a suitable simulation noise filtering rule. Experimental results covering a wide range of variations of the method under different congestion levels illustrate its advantageous performance.

2 Introduction

The traffic assignment problem amounts to solving a model system comprising a travel behavioral model that interacts with a network loading model. The travel behavioral model describes how travelers choose routes and possibly other aspects of travel (such as departure time or mode), and the network loading model describes the physical realization of the intended travel of possibly many travelers in a capacitated network. The *dynamic* traffic assignment (DTA) problem includes a within-day time dimension along which at least the network loading takes place. Wang et al. (2018) give ample references to concrete models. The arguably most intensively studied DTA problem class is that of deterministic continuum flow assignment. However, problem instances including realistic spatiotemporal flow dynamics have so far eluded a complete solution theory (Han et al., 2019), even though progress is being made under increasingly mild assumptions (Friesz et al., 2021).

The present article considers a *particle-based* and *stochastic* DTA problem. Particle-based means that the travel demand is composed of integral, decision making entities, and that the network loading is realized as a vehicle-discrete dynamic network flow simulator. Stochastic means that travel choices and/or network conditions

31 are random variables. Due to the intricate interactions between travel choices and net-
32 work flows, these distributions are in general not available in closed form; instead, the
33 model system only allows to simulate realizations thereof. This setup is representative
34 for many modern “agent-based” traffic assignment simulators and hence of increasing
35 practical relevance (Barcelo, 2010).

36 The considered class of DTA simulators can be characterized as process-based: a
37 scenario is composed of individual objects (travelers, vehicles, roads, ...), and the sce-
38 nario is evaluated by computing the interactions of these objects according to model-
39 based process rules. The possibility to freely select these processes renders them highly
40 versatile but also is the cause of their mathematical intractability. Different amounts
41 of information may be attached to a particle, and different travel behavioral degrees of
42 freedom may be considered.

- 43 • When assigning time-dependent origin/destination matrices, each particle be-
44 comes a trip maker (representing one trip in the matrix) that only is annotated
45 with origin, destination and departure time. Travel behavior is represented in
46 terms of a route, possibly a travel mode (car, public transport ...), and possibly a
47 departure time.
- 48 • When assigning a synthetic population (Farooq et al., 2013), each particle repre-
49 sents a synthetic traveler that is annotated with socio-demographic information
50 such as age, income or car ownership. Travel behavior is represented in terms of
51 an all-day travel plan that comprises a sequence of trips (annotated with route,
52 departure time, travel mode) connecting the traveler’s activity locations.

53 The following presentation considers the travel plan assignment of a synthetic pop-
54 ulation. This means that every particle represents a traveler, and every travel plan
55 represents the all-day travel (trip sequence) of that traveler. The route assignment of
56 an origin/destination matrix is a special case thereof. In reasonable generality, a DTA
57 simulation then functions according to Algorithm 1.

58 The present work considers instances of Algorithm 1 where each traveler computes
59 for each available plan a real-valued utility. The utility may depend on past network
60 conditions (it may, for instance, include travel time as a result of network congestion),
61 but given the traveler, the plan, and the past network conditions, utility is determinis-
62 tically determined. The specification is compatible with arbitrary random utility plan
63 choice models as long as the random utility terms are computed prior to the DTA
64 simulation and held constant throughout that simulation. This implies the behavioral
65 assumption that unobserved preferences and attributes of the alternatives do not vary
66 between assignment iterations.

67 It is assumed that travelers aim to maximize expected utility, but it is not assumed
68 that they dispose of an effective mechanism for doing so. It is assumed that the travel
69 behavioral model computes every assignment iteration for every traveler in a new *can-*
70 *didate* travel plan. The requirements on how this new plan is generated are very mild,
71 they range from a random variation of the previous plan to a best response plan that
72 would maximize the traveler’s expected utility given that all other travelers retain their
73 previously selected plans. Travelers want to switch to the candidate plan if this leads
74 to an expected utility gain. Informally (this is made concrete below), if no traveler

Algorithm 1 Blueprint of a DTA simulation

1. Represent the scenario. This comprises the transport network and the travel demand. The network may be multi-modal. The travel demand is a collection of individual travelers.
 2. Repeat the following process until a convergence criterion is satisfied (Nagel and Flötteröd, 2012).
 - (a) Every traveler chooses a travel plan. In a discrete choice framework, this step comprises both the possible update of a plan choice set and the choice of one plan from that set (Ben-Akiva and Lerman, 1985).
 - (b) All travelers are loaded onto the network. This means running a particle-based network flow simulator until all travelers have completed their daily travel. Traffic flow simulators come in many different guises (Barcelo, 2010).
 - (c) Every traveler processes the information gathered during the most recent network loading. This may be as simple as memorizing past travel times, possibly averaged over multiple past iterations (Cascetta, 1989).
 3. Extract solution statistics of interest. If variability in the process ceases and a point solution is attained, only the last iteration is of interest. If the process is stochastic and attains stationarity, one may compute means and variances over many stationary iterations.
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75 can increase expected utility by switching to the candidate plan, then the assignment
76 problem is considered solved.

77 DTA simulation practice has frequently resorted to variations of the method of suc-
78 cessive averages (MSA; Blum, 1954), which is well-known for guaranteeing conver-
79 gence when assigning certain classes of continuum flow models (Sheffi, 1985; Liu
80 et al., 2007). In the particle-discrete case, it can be used to control the replanning prob-
81 ability per traveler or the total number of replanners per assignment iteration; in either
82 case the frequency of replanning falls with one over the assignment iteration number.
83 The studies of Levin et al. (2015) and Ameli et al. (2020) compare DTA assignment
84 techniques including a range of MSA variations. Both present evidence that assigning
85 i.i.d. replanning probabilities to each particle is inferior to the “sorting technique” of
86 Sbayti et al. (2007) that replans in every assignment iteration the (MSA-controlled)
87 share of particles with the highest expected utility improvement.

88 Summing the expected utility improvement over all particles gives rise to a gap
89 function that is strictly non-negative and attains a zero value at the model’s solution
90 point. Mahut et al. (2008) and Himpe and Tampère (2016) discuss gap functions that
91 are applicable in dynamic and even particle-discrete settings. Lu et al. (2009) tackle the
92 assignment problem of Sbayti et al. (2007) within this setting. Smith (1984) adopts a
93 dynamical systems approach to the flow swapping process and minimizes a Lyapunov
94 (gap) function. Mounce and Carey (2011) analyze further developments of this ap-
95 proach. Most notable in the context of the present article is their interpretation of the
96 projection method of Wu et al. (1998) as a path flow swapping rule that aims to dampen
97 oscillations between iterations of the assignment process.

98 3 Method

99 Consider a population of $n = 1, \dots, N$ travelers. Let $\mathcal{X}_n$ be the set of all possible
100 travel plans available to traveler n , and let $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_N$ be the set of all
101 possible population travel plans. Denote by $x_n \in \mathcal{X}_n$ a concrete travel plan of traveler
102 n and by $x = [x_1, \dots, x_N] \in \mathcal{X}$ the fixed-size string of all concrete travel plans in the
103 population. The Hamming distance $d_H : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{N}_0$ between two population plans
104 $x, y \in \mathcal{X}$ is measured by counting the number of individuals with different plans in x
105 and y :

$$d_H(x, y) = \sum_{n=1}^N \mathbf{1}(x_n \neq y_n) \quad (1)$$

106 with $\mathbf{1}(\cdot)$ being the 0/1 indicator function. The population plans $\mathcal{X}$ are hence a subset
107 of a metric space.¹

108 Let $\mathcal{Z}$ be the set of all possible network conditions and let $Z(x) : \mathcal{X} \rightarrow \mathcal{Z}$ represent
109 the network conditions resulting from a (possibly stochastic) network loading of the

¹Extending the set of travel plans available to each agent to $\mathcal{X}_* = \bigcup_n \mathcal{X}_n$, a Hamming distance is defined over all size- N strings in $\mathcal{X}_* \times \dots \times \mathcal{X}_*$. The fact that each individual traveler n only considers a subset $\mathcal{X}_n \subseteq \mathcal{X}_*$ merely means that the distance function d_H exists over a larger domain ($\mathcal{X}_* \times \dots \times \mathcal{X}_*$) than what is considered here ($\mathcal{X}_1 \times \dots \times \mathcal{X}_N$).

population plans x onto the network. Let $v_n(r_n, z) : \mathcal{X}_n \times \mathcal{Z} \rightarrow \mathbb{R}$ be the real-valued utility receive by traveler n when executing travel plan $r_n \in \mathcal{X}_n$ in fixed network conditions z . Let

$$u_n(r_n, x) = E_{Z(x)}\{v_n(r_n, Z(x))\} \quad (2)$$

be the expected utility received by traveler n when executing the travel plan r_n over the distribution $Z(x)$ of network conditions that results from loading the population plans x onto the network. Let

$$u_n^*(x) = \max_{r_n \in \mathcal{X}_n} u_n(r_n, x) \quad (3)$$

be the largest expected utility individual n anticipates from unilaterally changing plans.

Given $x \in \mathcal{X}$, let x_{-n} denote x with x_n omitted, and let $[r_n, x_{-n}] \in \mathcal{X}$ denote the population plans that result from replacing x_n by r_n in the n th position of x and leaving the reminder of x (i.e. x_{-n}) unchanged.

Assumption 1. Let $x = [x_n, x_{-n}]$ and $y = [y_n, y_{-n}]$. There exist real-valued constants $C, D \geq 0$ such that the expected utility function u_n has the following properties, for all $n = 1, \dots, N$ and $x, y \in \mathcal{X}$:

$$|u_n(x_n, x) - u_n(x_n, y)| \leq C d_H(x, y) \quad (4)$$

$$|u_n^*(x) - u_n^*(y)| \leq D d_H(x, y) \quad (5)$$

where $|\cdot|$ denotes the absolute value function.

Assumption 1 resembles but deviates from that of Lipschitz continuity because d_H only takes integer values. This integrality implies that Assumption 1 is not stronger than assuming a finite value range of u_n : For $x = y$, both inequalities collapse into the expression $0 \leq 0$. If at least one plan differs between x and y , one has $d_H(x, y) \geq 1$ meaning that a sufficient condition to satisfy (4),(5) is to replace their right-hand sides by C resp. D , and then the existence of finite C, D is ensured by finite utility values on the left-hand sides. – Assumption 1 encodes the intuition that reducing the change in (population plans generating the) network conditions may also reduce the change in the expected utility of given plans, cf. (4), and of the utilities resulting from optimal reactions to the network conditions, cf. (5).

Let

$$g_n(r_n, x) = u_n(r_n, x) - u_n(x_n, x) \quad (6)$$

be the utility gain expected by traveler n from switching from plan x_n to plan r_n given that all other travelers keep their plans. The individual best-response gap $g_n^*(x) : \mathcal{X} \rightarrow \mathbb{R}$ and the population best-response gap $g^*(x) : \mathcal{X} \rightarrow \mathbb{R}$ are defined through

$$g_n^*(x) = u_n^*(x) - u_n(x_n, x) \quad (7)$$

$$g^*(x) = \sum_n g_n^*(x). \quad (8)$$

$g_n^*(x)$ measures the maximum expected utility gain that traveler n anticipates to achieve by unilaterally changing its travel plan. $g^*(x)$ is the corresponding population sum.

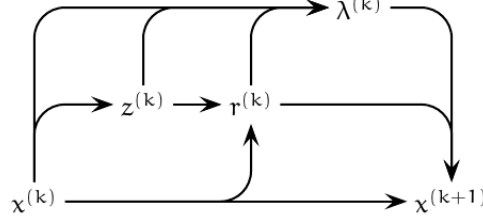


Figure 1: Stochastic process assignment

Definition 1. The population plans $x^{**} \in \mathcal{X}$ exactly solve the traffic assignment problem if the resulting population best response gap is zero:

$$g^*(x^{**}) = 0. \quad (9)$$

This means that no traveler anticipates an improvement of its expected utility from unilaterally switching to a different travel plan. The existence of such a solution depends on the concrete modeling assumptions encoded in the simulator. For broad applicability, and to bypass the need to validate the existence of a solution in the sense of Definition 1, the following, weaker formulation is adopted.

Definition 2. The population plans x^* optimally approximate a solution to the traffic assignment problem if the resulting expected population best response gap is minimal:

$$g^*(x^*) = \min_{x \in \mathcal{X}} g^*(x). \quad (10)$$

Since g^* is non-negative and attains zero at exact solutions, this gap minimization recovers exact solutions if they exist. An operational problem formulation is subsequently constructed that aims to minimize the expected population best response gap in the sense of Definition 2. From Assumption 1 follows

$$|g_n^*([x_n, x_{-n}]) - g_n^*([x_n, y_{-n}])| \leq (C + D)d_H(x_{-n}, y_{-n}), \quad (11)$$

which implies

$$g_n^*([x_n, x_{-n}]) \leq g_n^*([x_n, y_{-n}]) + (C + D)d_H(x_{-n}, y_{-n}). \quad (12)$$

The iterative traffic assignment process may now be considered as a time-inhomogeneous Markov chain with state space $\mathcal{X}$ and discrete time index k , cf. Figure 1: Given the population plans $x^{(k)}$ in assignment iteration k , a (possibly stochastic) network loading yields the realized network conditions $z^{(k)}$. The new population plans $x^{(k+1)}$ result from a subsequently detailed two-stage procedure of (i) receiving new candidate plans $r^{(k)} = (r_n^{(k)})$ for all travelers and (ii) deciding based on binary indicator variables $\lambda^{(k)} = (\lambda_n^{(k)})$ which of the new plans to take over into $x^{(k+1)}$.

Specifically, letting $r_n^{(k)} \in \mathcal{X}_n$ be a new plan computed by traveler n in iteration k , the following mechanism for generating $x^{(k+1)} = (x_n^{(k+1)})$ is postulated:

$$x_n^{(k+1)} = \begin{cases} r_n^{(k)} & \text{if } \lambda_n^{(k)} = 1 \\ x_n^{(k)} & \text{if } \lambda_n^{(k)} = 0 \end{cases} \quad (13)$$

164 with the replanning indicator $\lambda_n^{(k)} \in \{0, 1\}$. This means that traveler n keeps the old
 165 plan $x_n^{(k)}$ if $\lambda_n^{(k)} = 0$ and otherwise switches to $r_n^{(k)}$. Inserting (6) and (13) into (12)
 166 yields

$$g^*(x^{(k+1)}) \leq - \sum_n \lambda_n^{(k)} g_n(r_n^{(k)}, x^{(k)}) + (C + D)(N - 1) \sum_n \lambda_n^{(k)} + g^*(x^{(k)}). \quad (14)$$

167 The subsequently pursued assignment logic aims, in every iteration k of the assignment
 168 process, to select replanning indicators that minimize the upcoming iteration's gap
 169 $g^*(x^{(k+1)})$, i.e. the left-hand side of (14). Since $g^*(x^{(k+1)})$ is not yet known in
 170 iteration k , the approach is to reduce the right-hand-side of (14), which constitutes an
 171 upper bound on $g^*(x^{(k+1)})$.

172 Omitting the respect to the replanning indicators constant term $g^*(x^{(k)})$, a
 173 weighted sum of two terms remains on the right-hand side:

- 174 • $\sum_n \lambda_n^{(k)} g_n(r_n^{(k)}, x^{(k)})$ represents the expected utility gains of all plan-switching
 175 travelers. Its negative sign in (14) means that minimizing the right-hand side
 176 encourages replanning.
- 177 • $\sum_n \lambda_n^{(k)}$ is the total number of replanners. It is weighted with $(C + D)(N - 1)$
 178 and acts in (14) as a penalty term that discourages plan switching.

179 The parameters C and D are unknown. When estimating them, Assumption 1 is satis-
 180 fied as long as they are chosen sufficiently large. However, functioning as a replanning
 181 penalty weight, setting them to arbitrarily large values could impede all replanning.
 182 One may hence opt for gradually increasing estimators C and D over the assignment
 183 iterations k , expecting this to eventually reach values that comply with (4) while still
 184 allowing for replanning.

185 Observing that the only effect of increasing C and D is to reduce the number of
 186 replanners, this effect can just as well be achieved by explicitly limiting that number.
 187 This leads to the following formulation:

$$\begin{aligned} \min_{\lambda^{(k)} \in \{0, 1\}^N} \quad & - \sum_n \lambda_n^{(k)} g_n(r_n^{(k)}, x^{(k)}) \\ \text{s.t.} \quad & \sum_n \lambda_n^{(k)} \leq \Delta^{(k)} \end{aligned} \quad (15)$$

188 with $\Delta^{(k)}$ the maximum allowed number of replanners in assignment iteration k .

189 The problem (15) is solved by (i) sorting all travelers according to their expected
 190 gain $g_n(r_n^{(k)}, x^{(k)})$ and (ii) letting the $\Delta^{(k)}$ travelers with the largest (positive) gains
 191 replan. This logic coincides (apart from using the expected gap instead of a realization
 192 thereof) with the “sorting technique” of Sbayti et al. (2007). Beyond this, it has been
 193 derived from mild and operational mathematical assumptions.

194 Given the unavailability of closed-form distributional information, a numerical ap-
 195 proximation of the expected value $g_n(r_n^{(k)}, x^{(k)})$ in (15) is needed. With $z^{(k-r)}$ a re-

196 alization of $Z(x^{(k-r)})$ and $\mathcal{R} = \{0, \dots, R-1\}$, the following estimator is constructed:

$$\hat{u}_n(r_n, x; \mu, \mathcal{R}) = \sum_{r \in \mathcal{R}} \frac{\exp(-\mu d_H(x, x^{(k-r)}))}{\sum_{s \in \mathcal{R}} \exp(-\mu d_H(x, x^{(k-s)}))} v_n(r_n, z^{(k-r)}) \quad (16)$$

$$\hat{g}_n(r_n, x; \mu, \mathcal{R}) = \hat{u}_n(r_n, x; \mu, \mathcal{R}) - \hat{u}_n(x_n, x; \mu, \mathcal{R}). \quad (17)$$

197 This is a weighted sum of utility functions v_n , with the weights decaying exponentially
 198 with the distance between the population plans having generated the previous network
 199 condition realization (i.e. $x^{(k-r)}$ having generated $z^{(k-r)}$) and those plans for which
 200 an expectation is to be approximated (i.e. x). The function $\hat{g}_n(r_n, x)$ is a Kernel
 201 smoother over $\mathcal{X}$ that additionally is parameterized by $r_n \in \mathcal{X}_n$.

202 The parameter $\mu^{(k)}$ is estimated in every assignment iteration k as follows:

$$\mu^{(k)} = \arg \min_{\mu \geq 0} \sum_{n=0}^{N-1} \sum_{r \in \mathcal{R}} \left(v_n(r_n^{(k-r)}, z^{(k-r)}) - v_n(x_n^{(k-r)}, z^{(k-r)}) \dots \right. \\ \left. - \hat{g}_n(x_n^{(k-r)}, x^{(k-r)}; \mu, \mathcal{R} \setminus \{r\}) \right)^2. \quad (18)$$

203 Noting that $v_n(r_n^{(k-r)}, z^{(k-r)}) - v_n(x_n^{(k-r)}, z^{(k-r)})$ is the realization of a random
 204 variable with expectation $g_n(r_n^{(k-r)}, x^{(k-r)})$, this identifies a $\mu^{(k)}$ value that, on aver-
 205 age over all travelers n and past iterations $k-r$, reproduces this expectation as closely
 206 as possible. This is a leave-one-out estimator.²

207 4 Results and discussion

208 The above method development is the main result of this article. The present section
 209 experimentally illustrates this result. As a case study, the Greater Stockholm region
 210 in Sweden is considered. Figures 2(a) and (b) display the network. Only car traffic is
 211 considered; a 24-hour dynamic network loading is computed by a queueing simulation
 212 that approximates a particle-discrete kinematic wave model. The conflict resolution
 213 logic of its intersection model is stochastic. Travel behavioral degrees of freedom are
 214 route and departure time choice. Proposal routes are generated by computing time-
 215 dependent shortest paths against the previous assignment iteration; proposal departure
 216 times are generated by random variations.

217 We adopt from the recent study of Ameli et al. (2020) the following methods, which
 218 are applicable in the considered, particle based setting:

219 **I.i.d.** Given a replanning rate λ , each particle independently switches to its proposal
 220 plan with identical probability λ .

221 **Gap-prop.** Each particle computes the ratio of its current gap (i.e. expected cost re-
 222 duction) over the current cost (negative utility) of travel, multiplies this ratio

²If one evaluated $\hat{g}_n(x_n^{(k-r)}, x^{(k-r)}; \mu, \mathcal{R} \setminus \{r\})$ based on all past realizations $\mathcal{R}$, i.e. without excluding the lag r , then trivial solution $\mu = \infty$ would yield a zero objective function in (18) but not yield any smoothing over the simulation noise.

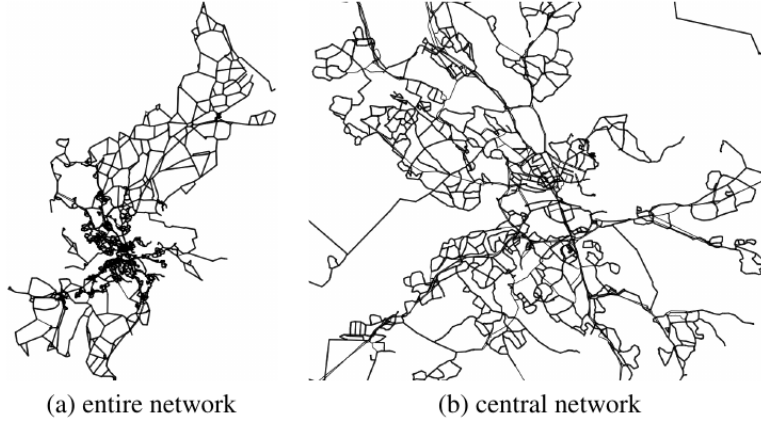


Figure 2: Stockholm network

with the replanning rate λ , and uses the resulting number as the probability of switching to its proposal plan.

Sbayti(mem) The proposed adaptation of Sbayti et al. (2007)’s algorithm, with “mem” representing the memory variable R that defines over how many averaging iterations the expected gap is approximated.

These methods are organized in the columns of Figure 3. All algorithms are evaluated for three different congestion levels: *low*, *medium* (the result of an approximate model calibration), and *high*. The congestion levels are organized in the rows of the following figures. All combinations of algorithms and congestion levels are evaluated for two different replanning rate schedules: *MSA* (λ falls with one over the iteration number) and *sqr*t (λ falls with one over the square root of the iteration number). A discussion of the *exp* schedule is omitted due to space restrictions. The schedules are differentiated by color coding. The following trends can be observed: Not surprisingly, gap levels increase with congestion levels. The *MSA* step size rule performs systematically unfavorably compared to the *sqr*t rule. *Sbayti* reaches lower gaps than *I.i.d.* and *Gap-prop*. Increasing the *Sbayti* memory from 1 to 10 iterations yields a clear improvement, but going beyond that and using a memory of 20 iterations has no clear advantage.

5 Conclusion

An exact treatment of the particle-based and stochastic traffic assignment problem is, without simplifying assumptions that would no longer cover existing simulator implementations, unavailable. Still, the increasing use of simulators in transport planning calls for further analysis. Given the wide spectrum of possible simulators, such an analysis can only be based on general, broad and hence rather simple assumptions. The present article delivers such an analysis for the widely used “sorting technique” of Sbayti et al. (2007) and demonstrates how such an analysis can support further al-

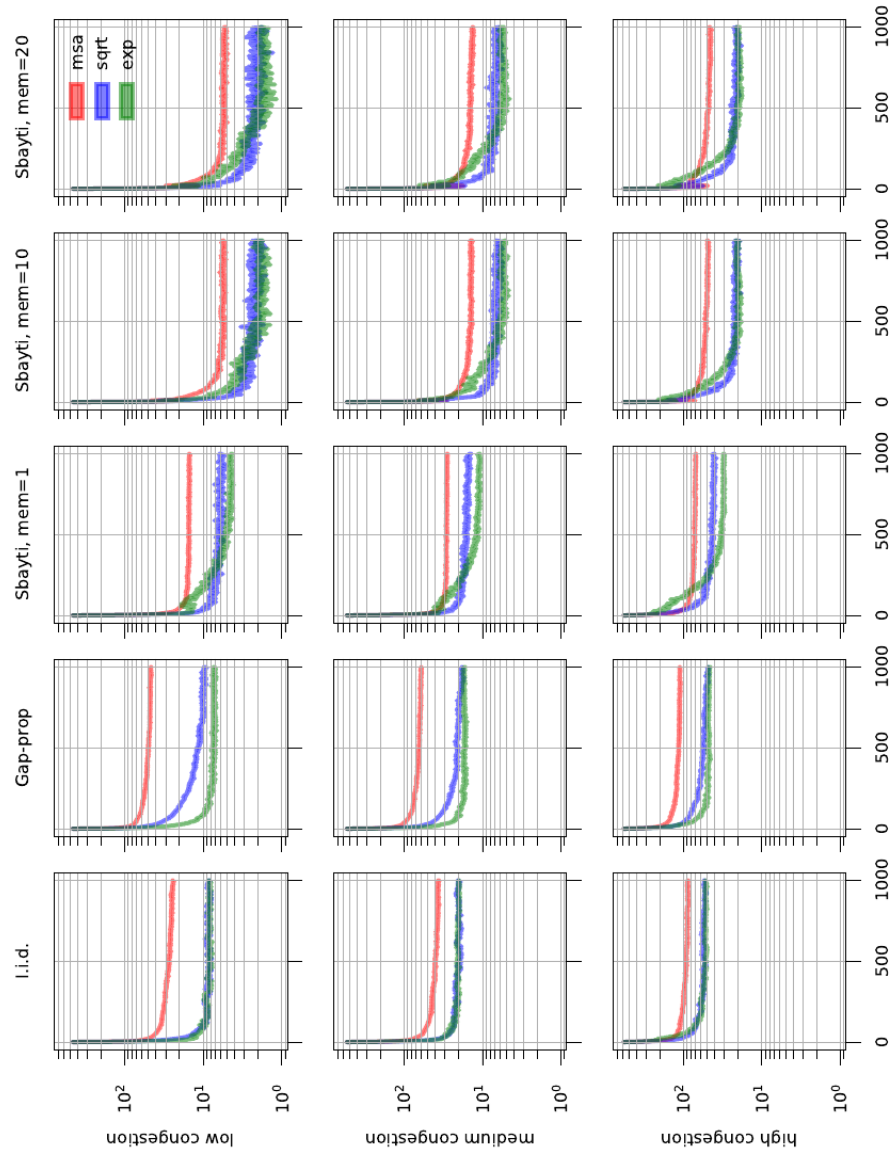


Figure 3: Gap over assignment iteration, 1000 iterations

248 gorithmic development. Experimental results indicate the practical relevance of the
249 approach.

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